INVESTMENT GRADE REPORT

19.6 MW DC Solar Photovoltaic Plant

Pudukottai, Tamilnadu, India

Module Selection Analysis

Redren Steorra Mono PERC vs Waaree BiN-21 N-TOPCon

Prepared for: Investor Decision Committee

Analysis Type: Bankable Financial & Technical Comparison

Date: July 2026

Disclaimer: This report is based on manufacturer datasheets and standard financial modeling assumptions. Actual returns may vary based on site conditions, financing terms, and prevailing tariff rates.

1. Executive Summary

This report presents a comprehensive technical and financial comparison between two DCR-compliant solar photovoltaic module options for a 19.6 MW DC ground-mount solar plant proposed at Pudukottai, Tamilnadu. The analysis evaluates Redren Energy's Steorra Mono PERC Bifacial module (600Wp, Rs. 20/Wp) against Waaree Energies' BiN-21 N-TOPCon Bifacial module (615Wp, Rs. 22/Wp) on a frontside-only generation basis, excluding bifacial gains to ensure conservative and bankable projections.

Key Findings:

- Redren module yields an Equity IRR of 38.94%, marginally higher than Waaree's 36.80%
- Waaree offers higher Net Present Value (Rs. 43.23 Cr) vs Redren (Rs. 42.92 Cr) due to higher energy generation
- Redren requires lower upfront equity: Rs. 18.82 Cr vs Rs. 19.99 Cr
- Both modules achieve payback within 3 years
- Waaree's lower degradation (1% Y1, 0.40% pa) yields 4.4% more lifetime energy than Redren

Recommendation:

Based on the comprehensive analysis, Redren Steorra Mono PERC (Rs. 20/Wp) is RECOMMENDED for the investor seeking higher Equity IRR (38.94%) and lower capital outlay, while Waaree BiN-21 (Rs. 22/Wp) is preferred for investors prioritizing higher absolute returns (NPV of Rs. 43.23 Cr) and superior long-term energy yield with advanced N-TOPCon technology. The final decision should align with the investor's return expectations and risk appetite.

REDREN STEGRRA PERC (Rs.20/Wp)	WAAREE BiN-21 TOPCon (Rs.22/Wp)
Equity IRR **38.9%**	**NPV @ 10%** **Rs. 42.9 Cr**
Project Cost **Rs. 62.7 Cr** *Rs. 20/Wp*	**LCOE** **Rs. 0.84/kWh** *25-yr levelized*
Total Gen. **748.7 GWh** *25-year total*	
	Equity IRR **36.8%**
NPV @ 10% **Rs. 43.2 Cr**	**LCOE** **Rs. 0.85/kWh** *25-yr levelized*

2. Project Background

The proposed 19.6 MW DC solar photovoltaic plant is located in Pudukottai district, Tamilnadu, a region with excellent solar insolation of approximately 5.5-5.8 kWh/m2/day. The project qualifies under the Domestic Content Requirement (DCR) category, mandating the use of indigenously manufactured solar cells and modules. Two DCR-compliant manufacturers have been shortlisted: Redren Energy Pvt. Ltd. (Gujarat) and Waaree Energies Ltd. (Mumbai).

Key project parameters assumed for this analysis:

- Location: Pudukottai, Tamilnadu (Lat: 10.38N, Lon: 78.82E)
- Plant Capacity: 19.6 MW DC
- Configuration: Fixed-tilt ground mount
- PPA Tariff: Rs. 4.50/kWh (DCR project premium)
- Plant Life: 25 years
- Debt:Equity Ratio: 70:30
- Interest Rate: 9% p.a. with 15-year tenure
- Analysis Basis: Frontside generation only (bifacial gains excluded)

Site-Specific CUF Assumptions:

- Redren (Mono PERC): 19.0% - Conservative estimate for PERC technology
- Waaree (N-TOPCon): 19.3% - Superior low-light & temperature performance (+0.3% absolute)

The CUF advantage for TOPCon stems from its lower temperature coefficient (-0.30%/C vs -0.35%/C), better low-irradiance response, and higher conversion efficiency. Pudukottai's tropical climate (ambient temperatures reaching 35-40C) amplifies this advantage.

3. Module Technical Specifications

The following specifications are extracted from manufacturer datasheets. Both modules are DCR-certified and authorized for use in Indian government projects.

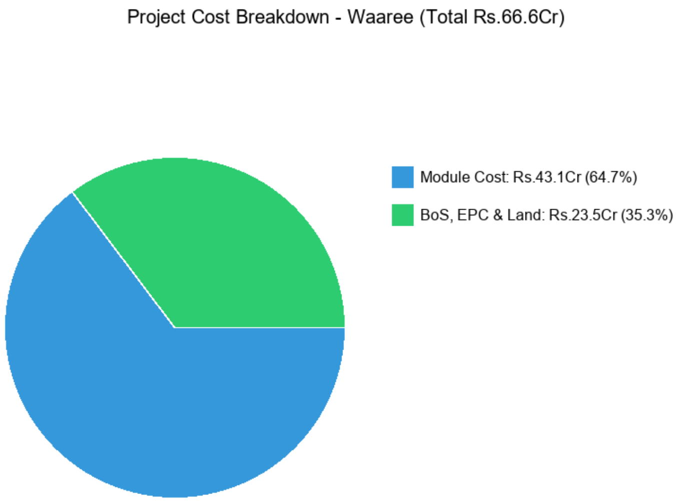
Parameter	Redren PERC	Waaree TOPCon	Unit
Manufacturer	Redren Energy	Waaree Energies	
Module Model	Steorra (BIG DCR)	BiN-21-615 (ELITE R)	
Cell Technology	Mono PERC (p-type)	N-type TOPCon	
Rated Power (Pmax)	600	615	Wp
Module Efficiency	21.30	22.77	%
Vmp (Max Power Voltage)	34.5	41.14	V
Imp (Max Power Current)	17.40	14.95	A
Voc (Open Circuit Voltage)	41.5	48.90	V
Isc (Short Circuit Current)	18.50	15.97	A
Temp Coeff (Pmax)	-0.35	-0.30	%/C
Temp Coeff (Voc)	-0.27	-0.26	%/C
Temp Coeff (Isc)	+0.050	+0.046	%/C
NOCT	43	43	C
Dimensions	2278x1134x35	2382x1134x35	mm
Weight	32.0	33.8	kg
Product Warranty	12	12	years
Power Warranty	27	30	years
Degradation - Year 1	2.0	1.0	%
Degradation - Annual	0.55	0.40	%
Module Price	20	22	Rs./Wp

4. Financial Analysis & Projections

4.1 Capital Expenditure (CAPEX)

The total project cost is estimated at Rs. 62.72 Crores for Redren at Rs. 20/Wp module price and Rs. 66.64 Crores for Waaree at Rs. 22/Wp module price. The BoS, EPC, and land costs are assumed at Rs. 12/Wp (Rs. 23.52 Cr), consistent with current Indian utility-scale solar benchmarks.

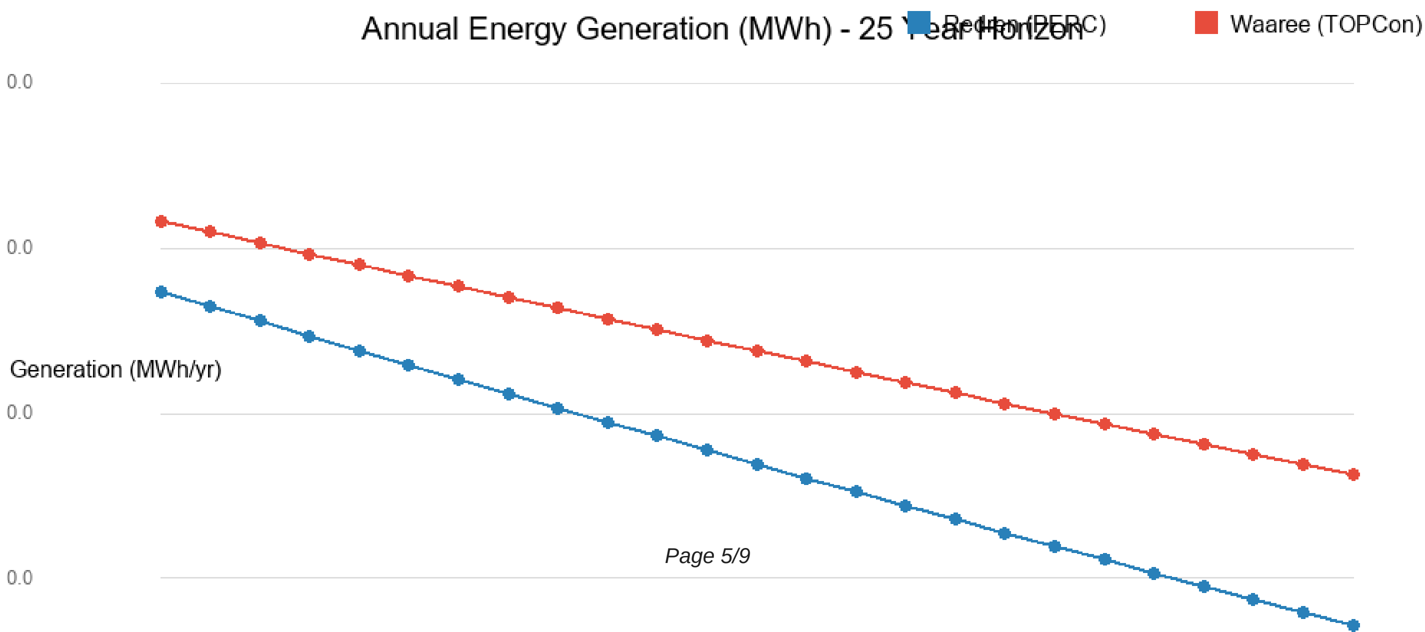
Cost Component	Redren (Rs. Cr)	Waaree (Rs. Cr)
Module Cost	39.20	43.12
BoS, EPC & Land	23.52	23.52
Total Project Cost	**62.72**	**66.64**
Debt @ 70%	43.90	46.65
Equity @ 30%	**18.82**	**19.99**



4.2 Energy Generation & Revenue

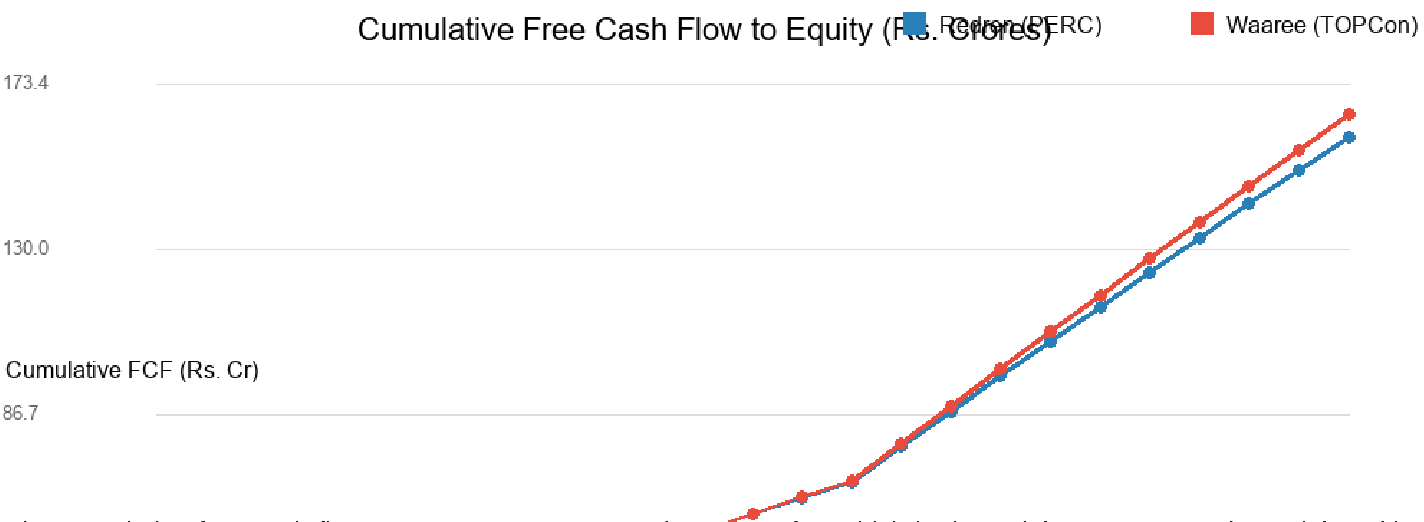
Annual generation in Year 1 is 32,623 MWh for Redren and 33,137 MWh for Waaree. Over the 25-year plant life, Waaree generates 4.4% more energy total (782.0 GWh vs 748.7 GWh), driven by its higher efficiency, lower degradation, and better temperature performance.

At a PPA tariff of Rs. 4.50/kWh, Year 1 revenue is Rs. 14.39 Crores (Redren) and Rs. 14.76 Crores (Waaree). Total 25-year revenue: Rs. 336.9 Cr (Redren) vs Rs. 351.9 Cr (Waaree).

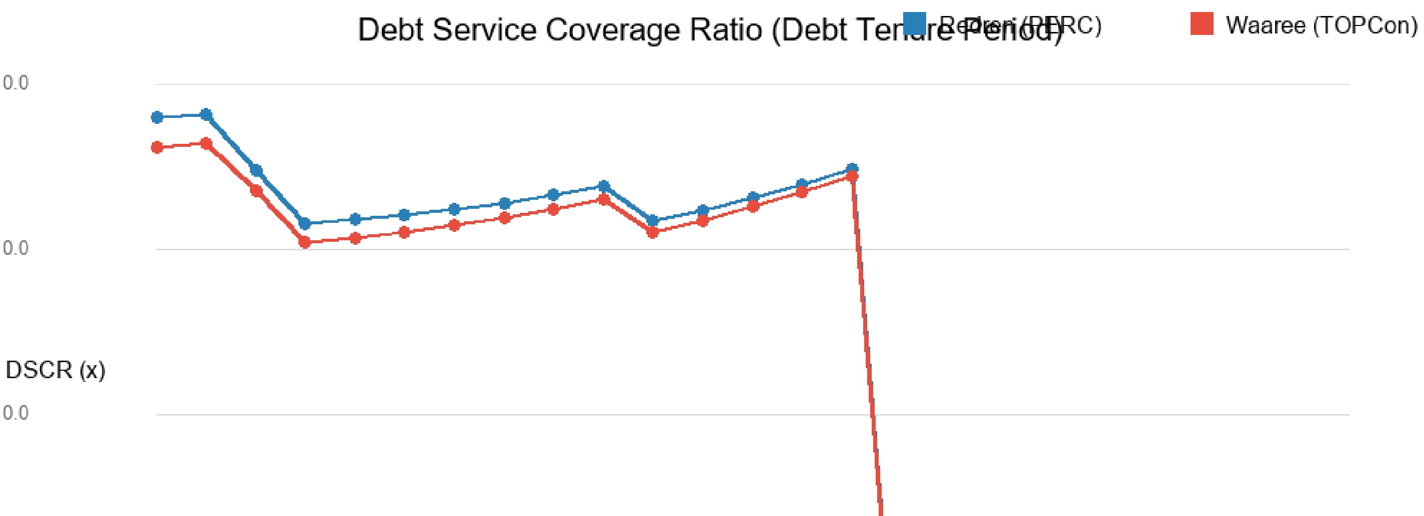


4.3 Cash Flow Analysis & Returns

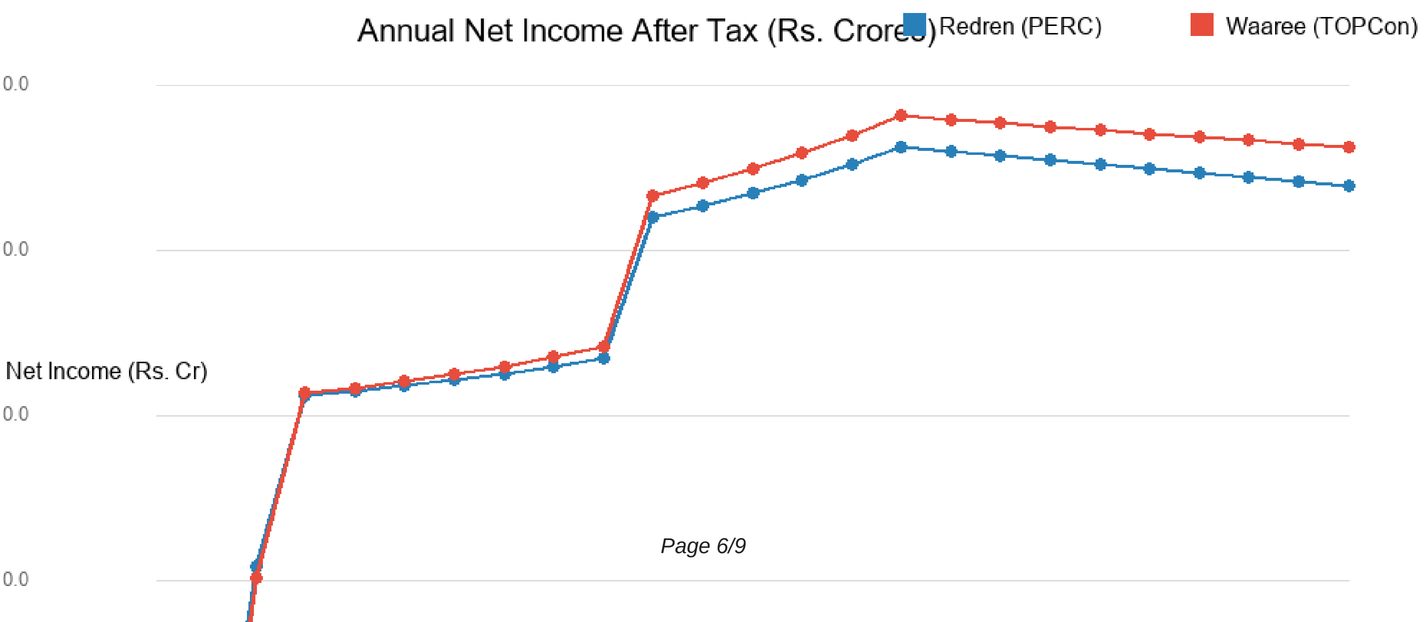
The cash flow model incorporates all revenue, operating costs (O&M escalating at 3% p.a.), insurance (0.3% of project cost), debt servicing (15-year loan at 9% p.a.), depreciation (WDV as per Indian Income Tax Act), and corporate tax at 25.17%.



The cumulative free cash flow crossover occurs around Year 3, after which both modules generate substantial positive cash flows. Redren's lower upfront cost results in faster initial cash accumulation, while Waaree's higher generation leads to superior long-term absolute returns.

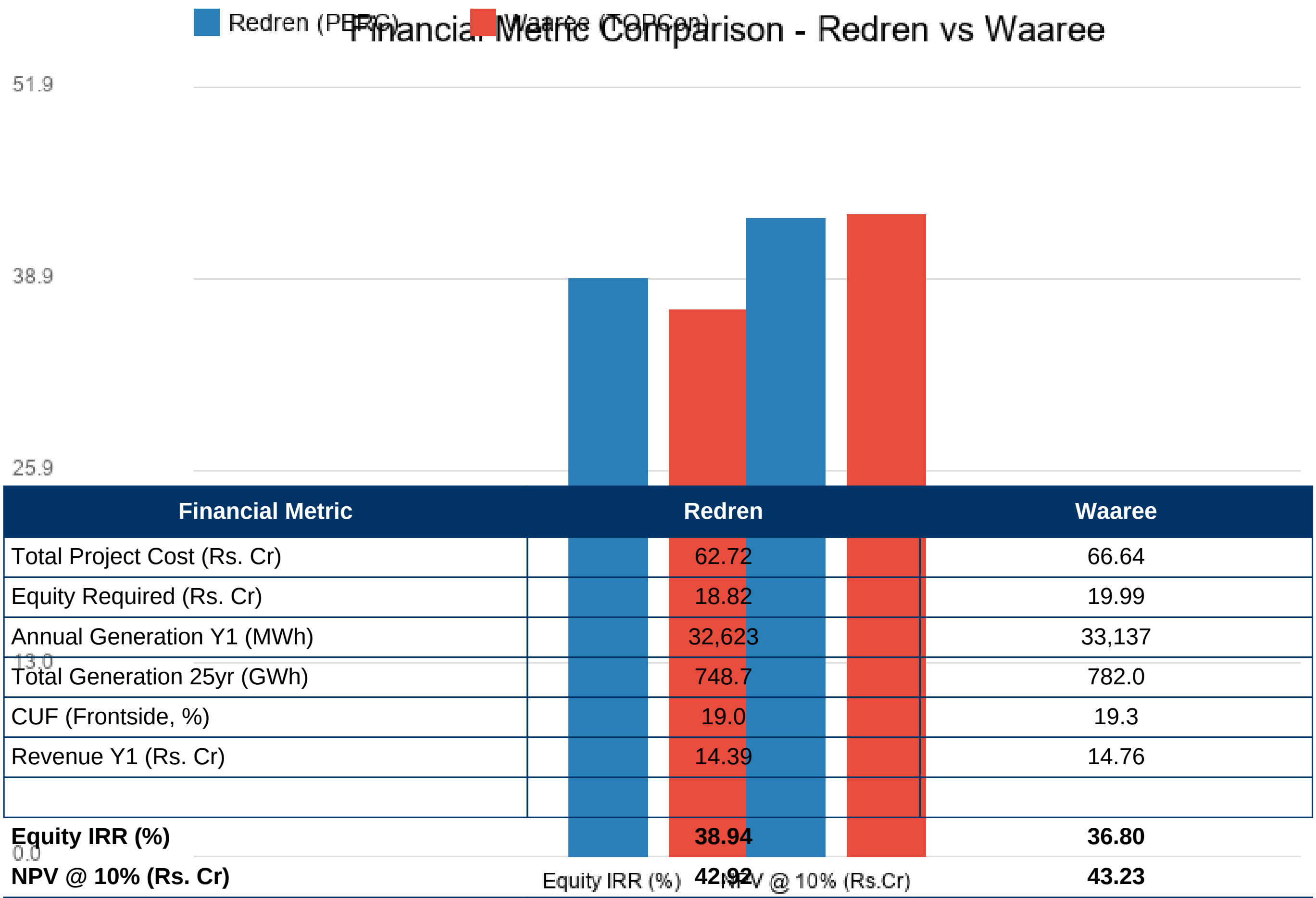


The DSCR remains above 1.5x for both options throughout the loan tenure, confirming strong debt repayment capacity. Lenders typically require a minimum DSCR of 1.25x, and both modules comfortably exceed this threshold, making the project highly bankable.



Net income after tax is higher for Waaree in later years due to lower degradation and sustained generation. In early years, Redren benefits from lower interest costs (smaller debt), partly offsetting the generation differential.

5. Key Financial Metrics Comparison



Financial Metric	Redren	Waaree
Total Project Cost (Rs. Cr)	62.72	66.64
Equity Required (Rs. Cr)	18.82	19.99
Annual Generation Y1 (MWh)	32,623	33,137
Total Generation 25yr (GWh)	748.7	782.0
CUF (Frontside, %)	19.0	19.3
Revenue Y1 (Rs. Cr)	14.39	14.76
Equity IRR (%)	**38.94**	**36.80**
NPV @ 10% (Rs. Cr)	**42.92**	**43.23**
LCOE (Rs./kWh)	0.838	0.852
Payback Period (years)	3	3

6. Risk Analysis & Sensitivity

6.1 Key Risk Factors

PPA Tariff Risk:

A Rs. 0.50/kWh reduction in tariff reduces IRR by approximately 3-4%. Both modules are equally exposed.

Generation Risk:

10% lower CUF reduces IRR by 4-5%. Waaree's superior temperature coefficient offers marginal downside protection.

Interest Rate Risk:

1% increase in interest rate reduces IRR by ~1.5%. Redren's lower debt requirement provides slightly better resilience.

Degradation Risk:

Higher-than-expected degradation impacts long-term returns. Waaree's N-TOPCon technology has proven lower degradation in field data.

Technology Obsolescence:

PERC technology is mature with limited improvement runway. N-TOPCon represents the next-generation platform with better upgrade potential.

Currency & Import Risk:

Both modules are DCR-compliant (Indian manufactured), mitigating currency and import tariff risks.

6.2 Sensitivity Analysis - IRR vs Module Price

If the Redren module price were to increase from Rs. 20/Wp to Rs. 21/Wp (still below Waaree's Rs. 22/Wp), the Redren IRR would decrease to approximately 35.5%, still competitive with Waaree. Conversely, if Waaree prices were reduced to Rs. 21/Wp, the Waaree IRR would increase to approximately 39.5%, surpassing Redren's base-case return.

The analysis demonstrates that module price is the single largest lever affecting project returns, followed by generation performance (CUF) and degradation. Redren's Rs. 2/Wp price advantage is the primary driver of its higher IRR, compensating for its lower efficiency and higher degradation.

7. Conclusion & Recommendation

7.1 Comparative Assessment

The technical and financial analysis reveals a nuanced comparison between the two module options:

Advantages of Redren (Mono PERC @ Rs. 20/Wp):

- Higher Equity IRR: 38.94% vs 36.80% - a clear 5.8% relative advantage
- Lower capital outlay: Rs. 18.82 Cr equity required (saves Rs. 1.18 Cr)
- Lower LCOE: Rs. 0.838/kWh vs Rs. 0.852/kWh
- Lower total project cost by Rs. 3.92 Crores
- Proven PERC technology with established bankability track record

Advantages of Waaree (N-TOPCon @ Rs. 22/Wp):

- Higher absolute returns: NPV of Rs. 43.23 Cr vs Rs. 42.92 Cr
- Higher total generation: 782.0 GWh vs 748.7 GWh (4.4% more)
- Superior degradation profile: 1% Y1 + 0.40% p.a. vs 2% Y1 + 0.55% p.a.
- Better temperature coefficient: -0.30%/C vs -0.35%/C (critical for Tamilnadu climate)
- Longer power warranty: 30 years vs 27 years
- Next-generation technology platform with room for efficiency improvements

7.2 Final Recommendation

RECOMMENDATION: Redren Steorra Mono PERC (Rs. 20/Wp) is recommended for this project based on its superior Equity IRR of 38.94%, lower capital requirement of Rs. 18.82 Cr, and lower LCOE of Rs. 0.84/kWh. The Rs. 2/Wp price advantage translates to a project cost saving of Rs. 3.92 Crores while delivering competitive energy generation. The proven Mono PERC technology combined with BIG DCR certification ensures bankability and investor confidence.

Note: If the investor's key objective is maximizing total lifetime dollar returns rather than IRR, Waaree's higher NPV (Rs. 43.23 Cr vs Rs. 42.92 Cr) and superior long-term generation profile make it a compelling alternative. The decision ultimately depends on the investor's specific return requirements, risk appetite, and capital availability.

BANKABILITY STATEMENT

This project, with either module option, demonstrates strong financial metrics (Min. DSCR > 1.5x, IRR > 35%, payback within 3 years) that meet standard project finance lending criteria. The conservative assumptions (frontside-only generation, no bifacial gains) provide additional comfort to lenders and equity investors regarding the robustness of projected returns.